

In a single network configuration, all Matter Nodes are attached to the same IPv6 link. A single network configuration may consist of a single bridged Wi-Fi / Ethernet network where all nodes attached to that network are part of the same broadcast domain. When all Matter Nodes are attached to the same Wi-Fi / Ethernet network, link-local IPv6 addressing is sufficient - no additional IPv6 network infrastructure is required.

In a multiple network configuration, a Matter network is composed of (typically one) infrastructure network and one or more stub networks. Unlike an infrastructure network, stub networks do not serve as transit networks. Typically, the infrastructure network is a bridged Wi-Fi / Ethernet network and the Thread networks are stub networks. A stub router connects a stub network to an infrastructure network and provides IPv6 reachability between the two networks.

4.2.1. Stub Router Behavior

A stub router SHALL implement [\[draft-lemon-stub-networks\]](#). In a multiple network configuration, both the infrastructure and stub networks require routable IPv6 addresses to communicate across networks. A routable IPv6 address SHALL have global scope (e.g. GUA or ULA) [\[RFC 4007\]](#) and SHALL be constructed out of a prefix advertised as on-link. If there is no routable prefix on a given network, the stub router SHALL provide its own routable prefix. Note that Thread's "on-mesh prefix" is equivalent to Wi-Fi / Ethernet's "on-link prefix".

Stub routers SHALL advertise reachability to all routable prefixes on the adjacent network. A stub router connecting a Thread network SHALL advertise reachability to all of the Thread network's routable prefixes to the adjacent infrastructure network using Route Information Options [\[RFC 4191\]](#) contained in Router Advertisements [\[RFC 4861\]](#). That same stub router SHALL also advertise reachability to all of the infrastructure network's routable prefixes to the adjacent Thread network in the Thread Network Data [\[Thread specification\]](#).

4.2.2. Matter Node Behavior

Matter Nodes SHALL configure a link-local IPv6 address. In addition, Nodes SHALL support configuration of at least three routable IPv6 addresses (in addition to the link-local and, in the case of Thread, mesh-local addresses). On a Wi-Fi / Ethernet interface, ICMPv6 Router Advertisement (RA) messages advertise prefixes for use on the link [\[RFC 4861\]](#). On a Thread interface, the Thread Network Data contains prefixes for use on the link [\[Thread specification\]](#). If a Node receives an on-link prefix that allows autonomous configuration on a given interface and the Node has fewer than three routable IPv6 addresses configured, the Node SHALL autonomously configure an IPv6 address out of that prefix.

Matter Nodes SHALL also configure routes to adjacent networks. On Wi-Fi / Ethernet networks, Nodes SHALL process Route Information Options [\[RFC 4191\]](#) and configure routes to IPv6 prefixes assigned to stub networks via stub routers. Wi-Fi / Ethernet interfaces SHALL support maintaining at least 16 different routes configured using Route Information Options. On Thread networks, Nodes SHALL route according to routing information provided in the Thread Network Data [\[Thread specification\]](#). Thread devices SHALL support as many routes as can be encoded in the Thread Network Data.

Matter Nodes SHALL support a number of IPv6 neighbor cache entries at least as large as the [number of supported CASE sessions](#) plus the number of supported routes.